

Embeddings and BERTopic



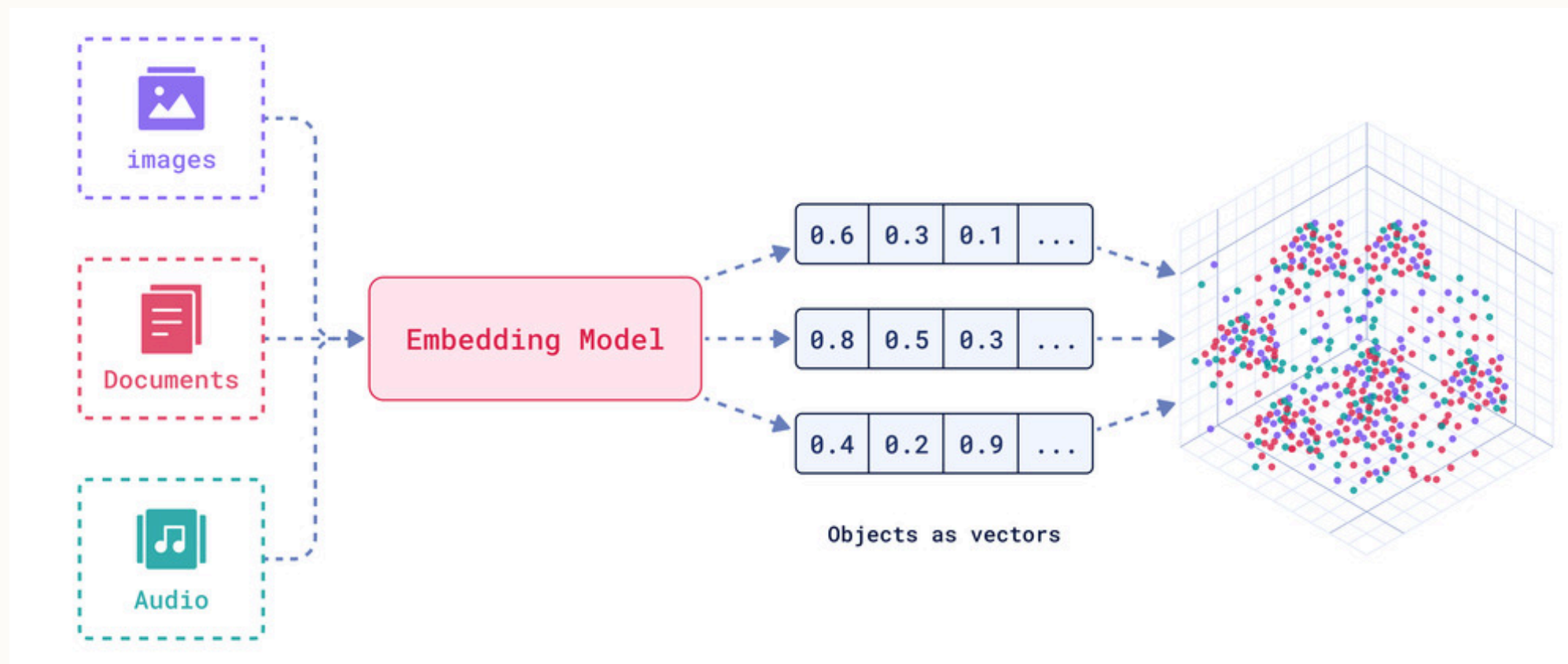
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Embeddings

turning text into number



vectors as GPS location for text

not only latitude and longitude, but 1000+ numbers
(e.g. word "Apple" [0.98, 0.05, 0.92, 0.02, 0.10,...])

location can change

not due to force, but thanks to neighbours
(e.g. Apple between Microsoft and Google will have a different location than Apple between Banana and Orange)

you can measure distance

not with ruler, but with cosine similarity or distance (0-1)
(e.g. similarity between "Apple" and "Blueberry" will be bigger 0.8, between "Apple" and "Guitar", similarity will be smaller 0.5)

you can use average

not a city centre, but centroid
(e.g. vector location of "Apple" + vector location of "Bananna" + vector location of "Blueberry" / 3 = location between them)

Embedding models

static models - each word has a fixed vector (Word2Vec)

dynamic models - different vector based on other words (transformer architecture)

closed source (e.g. OpenAI's models) - costs, might change, data privacy

open source (e.g. BERT) - slow, not as good

general-purpose models (e.g. BERT)

domain-specific models (e.g. Legal-BERT)

Embeddings

once you can turn text into number, you can..

1. Compare questionnaires with each other or items with their intended measured symptom
2. Handle jingle (same words, different meaning) or jangle (different words, same meaning) fallacy
3. Predict correlations in human data based on the semantic similarity of items
4. Verify the factor structure using semantic meaning before finalizing a questionnaire
5. Improve questionnaires by shortening scales and ensuring each question is unique
6. Extract measurable psychological variables from unstructured text data

... and much more



Google Colab

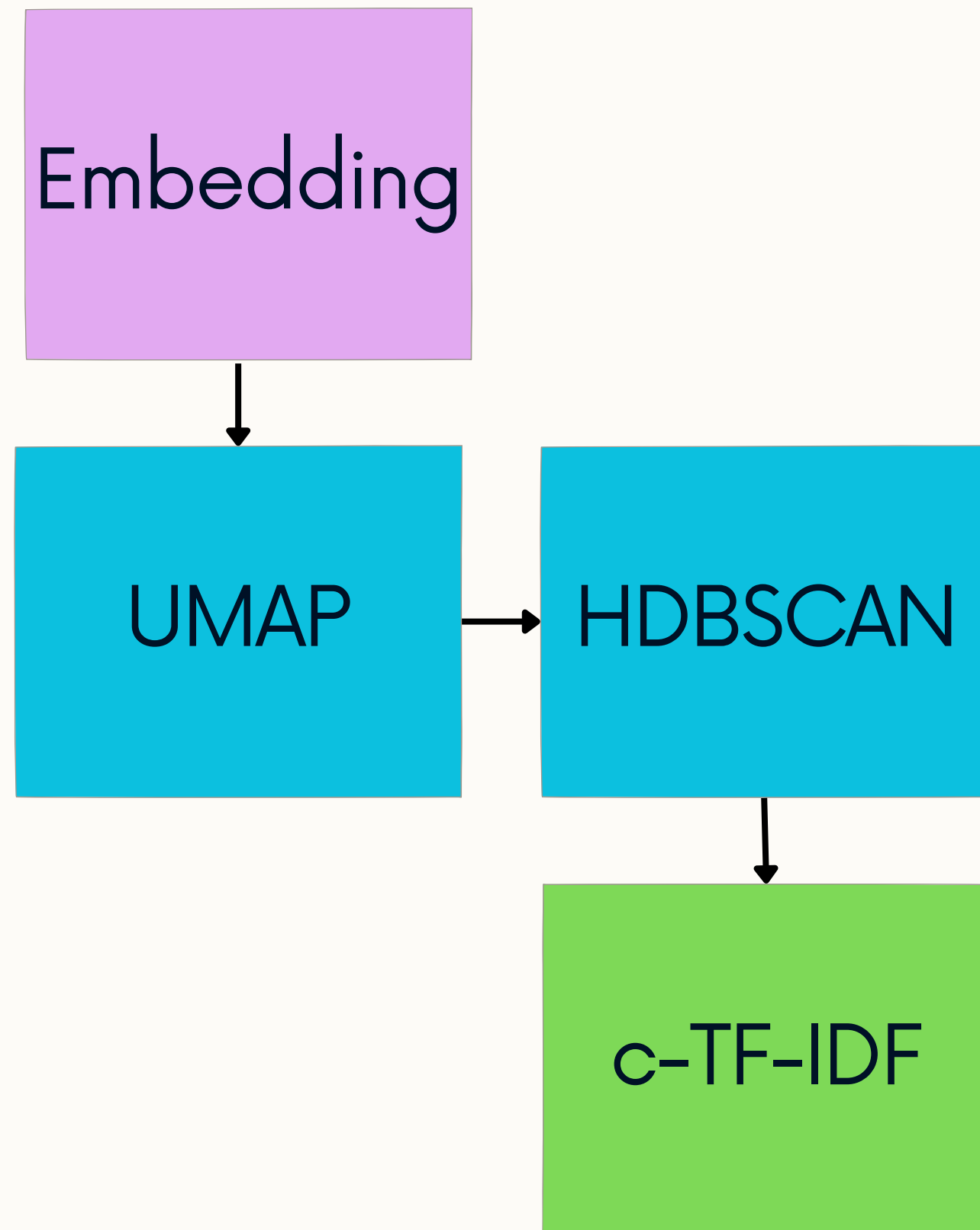
[google.com](https://colab.google.com)

What can BERTopic solve?

- Analyzing large datasets (abstracts, social media posts, open-ended questions,...)
- Identifying latent themes – colored space
- Identifying gaps – white space
- Measuring novelty – against corpus of papers via embedding
- Longitudinal trend analysis
- Structured categorization – clustering and hierarchization
- Linking it with other data

Maarten Grootendorst is the creator of BERTopic detailed it in a 2022 paper titled "BERTopic: Neural topic modeling with a class-based TF-IDF procedure". Trained psychologist, then data scientist, currently working in DeepMind.

What is BERTopic?



1.Document Embedding – Convert text to dense vectors (numerical representation) capturing context using language model (either BERT, or recent LLMs, which projects each abstract into a 3,072-dimensional semantic space)

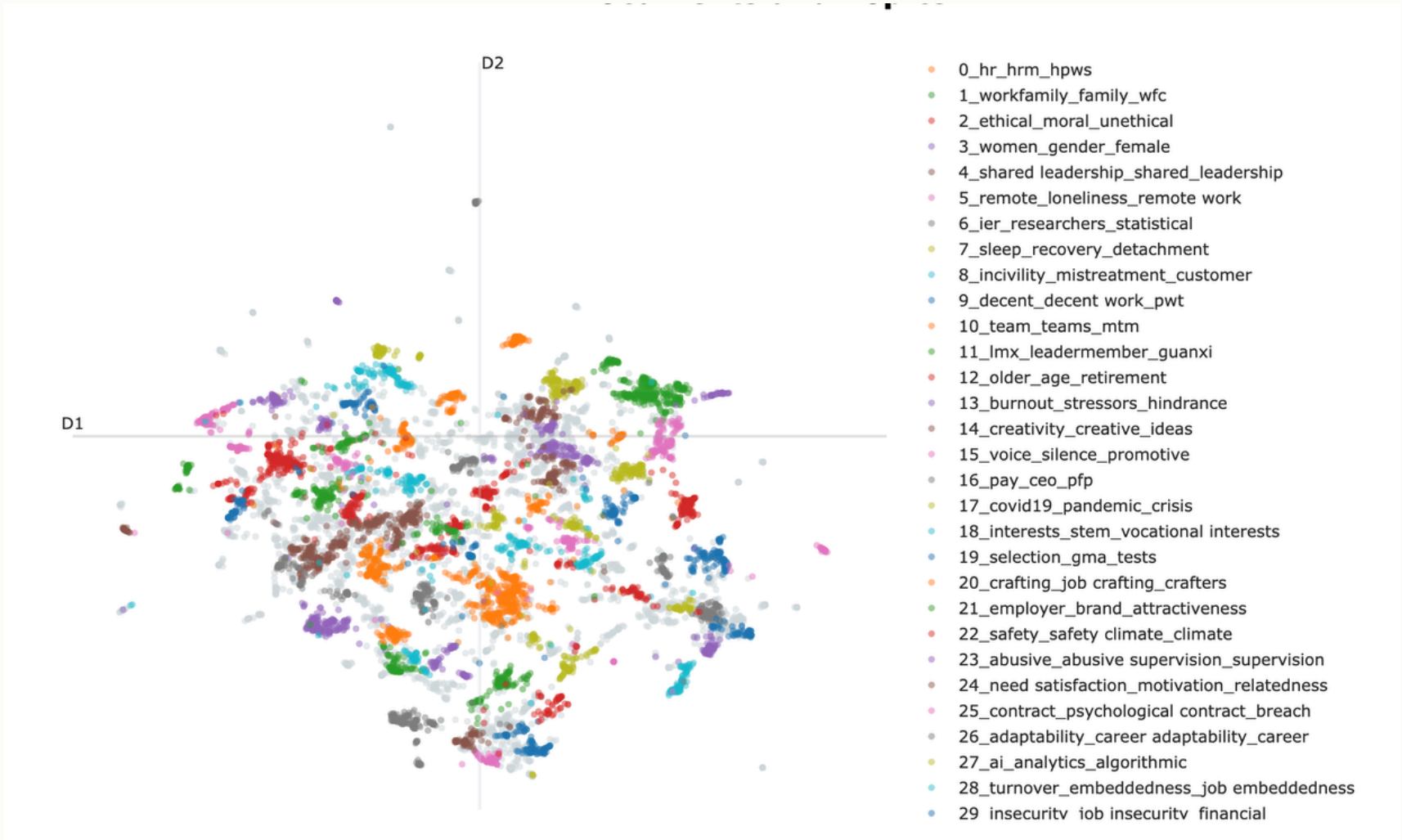
2.Dimensionality Reduction – BERTopic standardly uses Uniform Manifold Approximation and Projection (UMAP) to reduce these embeddings into a lower-dimensional space while preserving structure of the data.

3.Document Clustering – Algorithm groups semantically similar documents together. It typically employs Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). You can use „stop-words“ and adjust the number and size of clusters. Able to identify „noise“ and outliers.

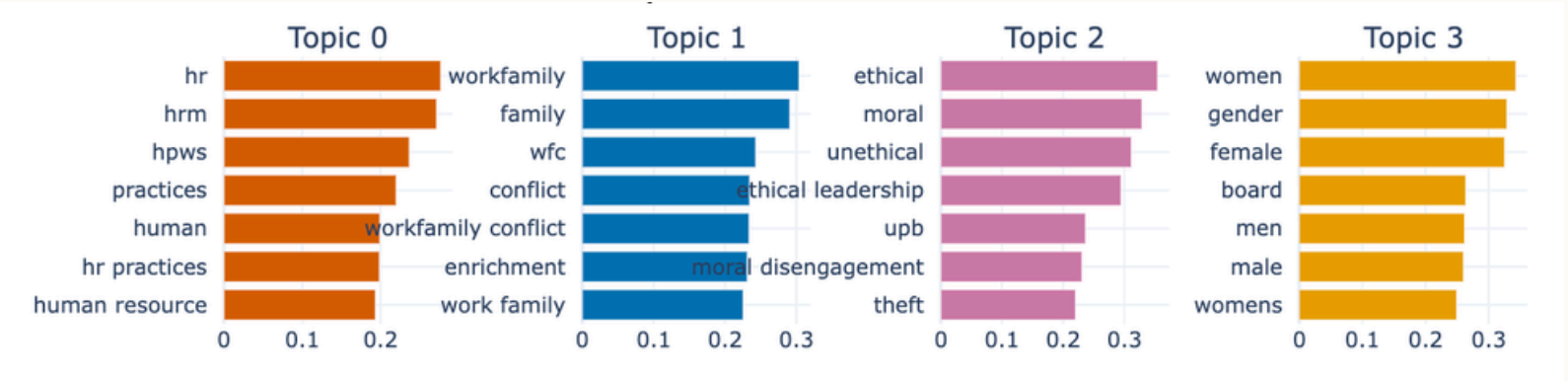
4.Topic Representation – BERTopic treats each cluster as a single large document and applies a custom c-TF-IDF procedure to extract the most representative words for each cluster to define the topic.

Outcomes of BERTopic

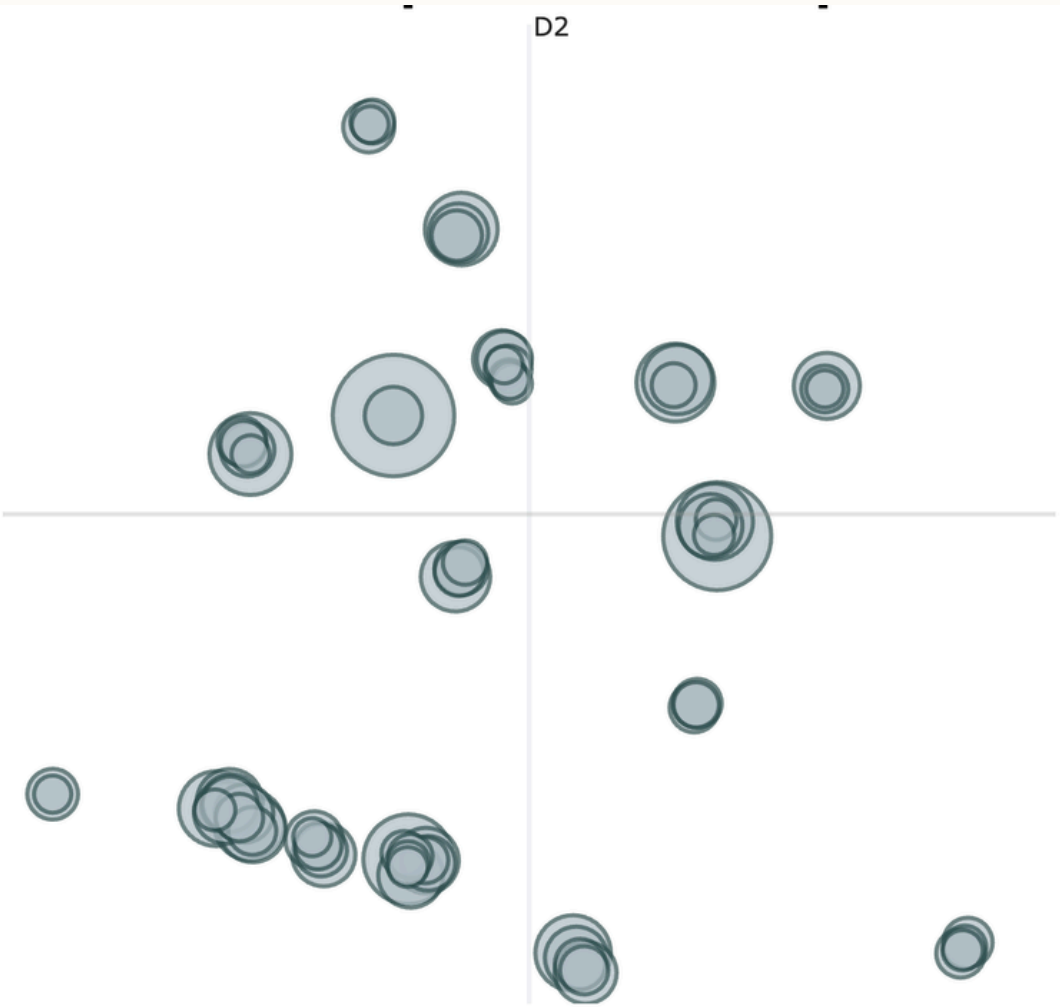
Uniform Manifold Approximation and Projection



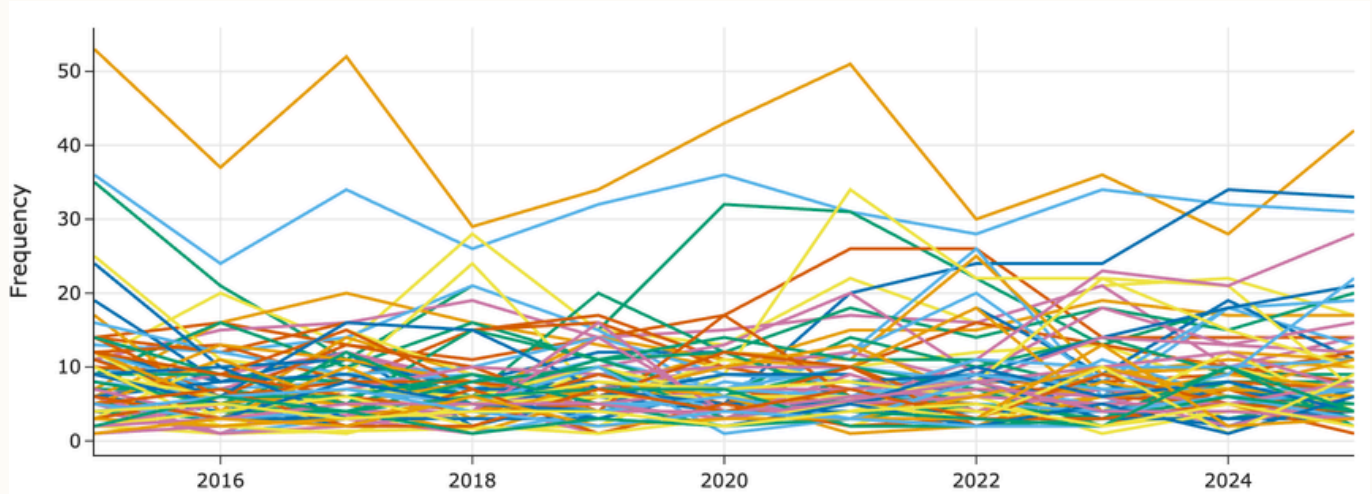
Topic Representations

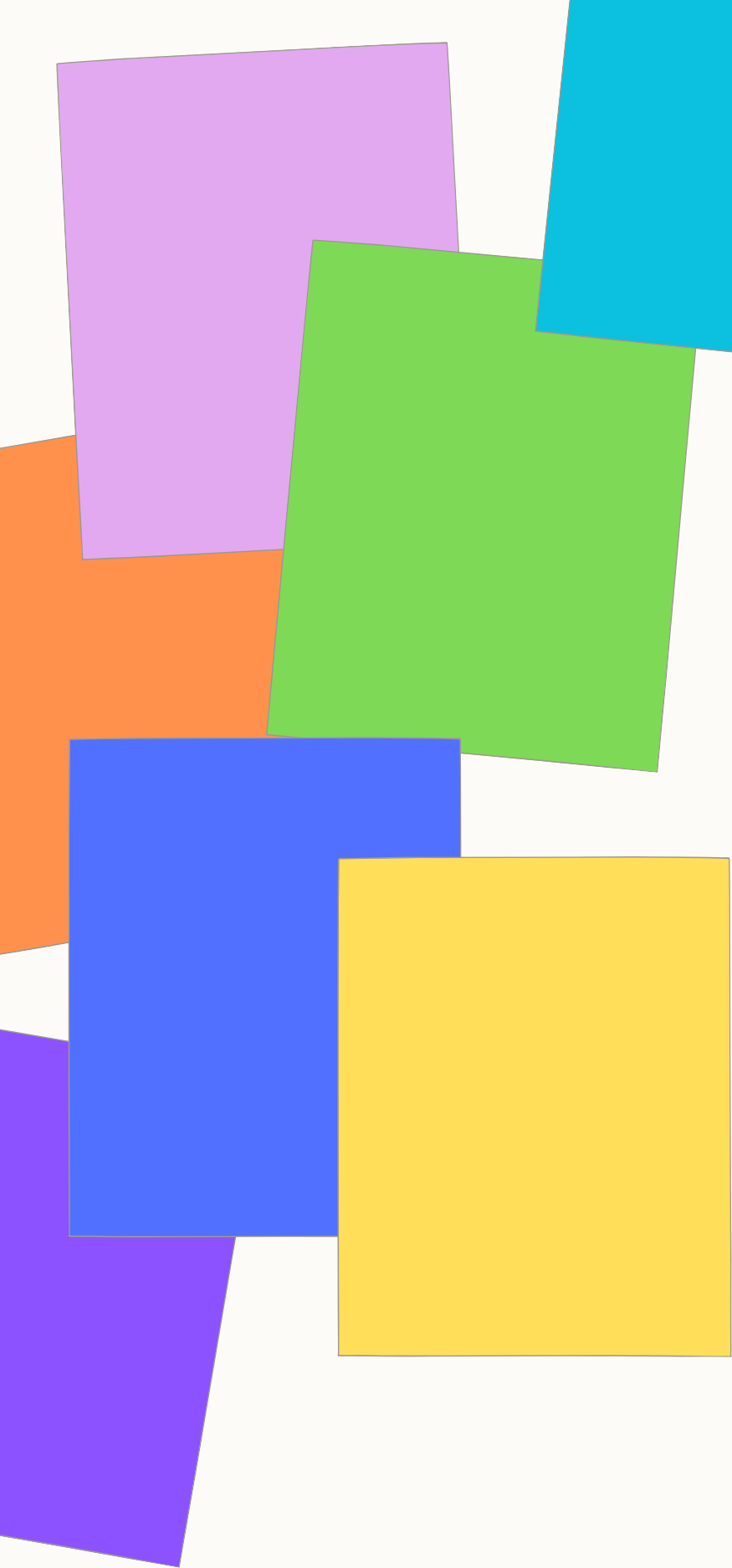


Intertopic Distance Map



Dynamic Topic Modeling





Metrics

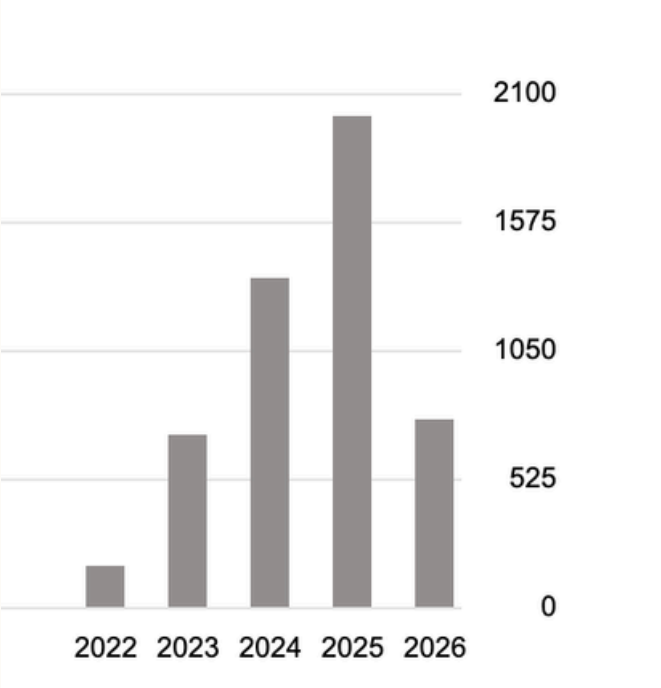
- Topic Coherence using NPMI – evaluates the semantic interpretability of the topics.
- Topic Diversity – calculates the percentage of unique words.
- Computation time – duration (in seconds) required to train the model.
- Silhouette Score – how similar an object is to its own cluster compared to other clusters
- KL-Uniform – how much a topic's word distribution differs from generic distribution.
- Human Evaluation – e.g. Likert Scale to rate topic's (in)coherence

Limits

- Each document contains only a single topic.
- Assigns a high percentage of documents to an outlier cluster.
- While the initial embedding is contextual, the final c-TF-IDF representation is still based on a frequency
- The quality of the topics is heavily dependent on the embedding model chosen.
- Lack of standardized objective metrics.

Gaining popularity

Feuerriegel, S., Maarouf, A., Bär, D. et al. Using natural language processing to analyse text data in behavioural science. **Nat Rev Psychol** 4, 96–111 (2025). <https://doi.org/10.1038/s44159-024-00392-z>
highlight BERTopic for topic modeling



<https://arxiv.org/abs/2203.05794>

Topic Evolution in Positive **Psychology** and Schizophrenia Research: A **BERTopic**-Based Trend Analysis (2015–2025)

[YC Lin](#), [KJ Huang](#) - 2026 - [researchsquare.com](#)

... A semantic embedding-based **BERTopic** model, an advanced artificial intelligence tool for ... bio-**psychological** integration, defined as the convergence of biological and **psychological** ...

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BERT and **BERTopic** for screening clinical depression on open-ended text messages collected through a mobile application from older adults

[MK Chung](#), [SY Lee](#), [T Shin](#), [JY Park](#), [S Hwang](#)... - [BMC public health](#), 2025 - [Springer](#)

... has the benefit of revealing their **psychological** state without the pressure of taking a survey ... depression without triggering significant defensive **psychological** reactions, especially from ...

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Mapping the landscape of AI-empowered **psychology**: a topic modeling-based bibliometric analysis

[S Jia](#), [Y Zhang](#), [F Wang](#) - [Artificial Intelligence Review](#), 2026 - [Springer](#)

... (AI) is profoundly reshaping **psychological** research and ... **psychology**" to delineate its thematic structure, developmental trajectory, and publication patterns. We conducted a **BERTopic**-...

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[HTML] Strategic Management Knowledge Map via **BERTopic** (1980–2025): Evolution, Integration, and Application

[KK Lai](#), [CW Hsiao](#), [YJ Hsu](#) - [Applied System Innovation](#), 2025 - [mdpi.com](#)

... These developments broaden sources of insight across economics, sociology, **psychology**, and management and blur theoretical boundaries, leading to paradigm expansion, cross-...

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Trends and thematic evolutions in **psychological** first aid research in disasters: insights from bibliometric and **BERTopic** analysis research

[G Sen](#) - [Current Psychology](#), 2026 - [Springer](#)

By combining bibliometric mapping and transformer-based topic modeling (**BERTopic**), this study offers an in-depth analysis of trends and thematic developments in PsychologicalFirst ...

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Exploiting topic analysis models to explore **psychological** dimensions in social media data

[M Couto](#), [J Parapar](#), [DE Losada](#) - [Scientific Reports](#), 2026 - [nature.com](#)

... datasets where thematic content does not always reveal **psychological** cues. Our results demonstrate that **BERTopic** significantly outperforms other models in perceived coherence, ...

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